

Adham Ashraf Saeed

Assignment 4 (Final Project)

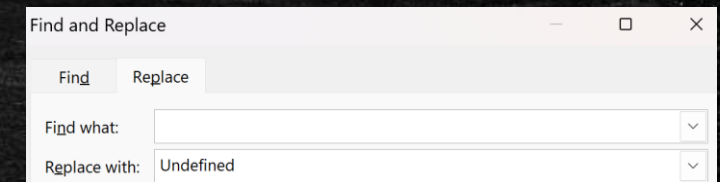
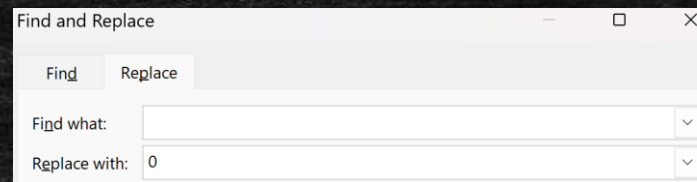
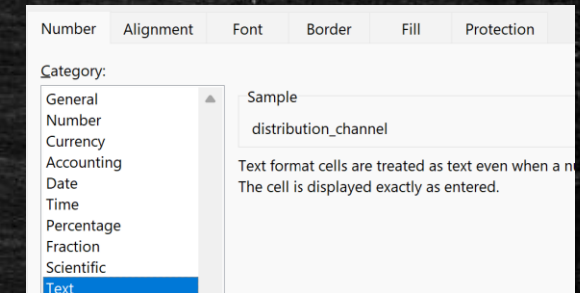
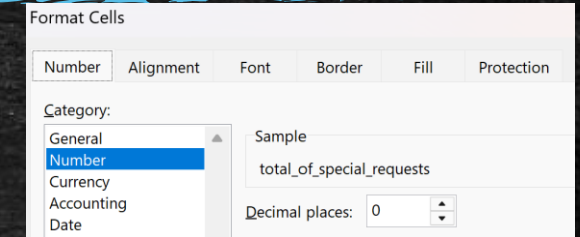
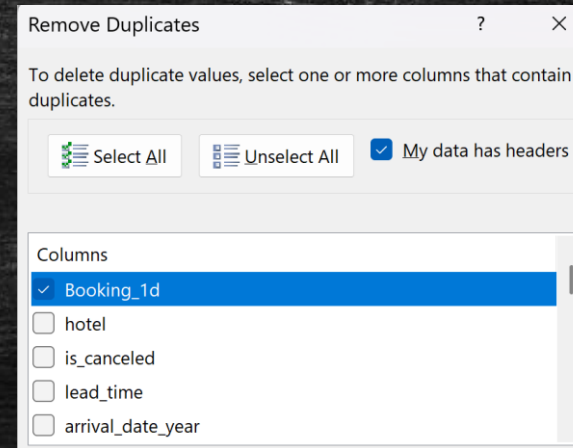
Project Objectives

- Clean and prepare raw data
- Use Excel formulas effectively
- Perform exploratory data analysis
- Build Pivot Tables & dashboards
- Generate business insights

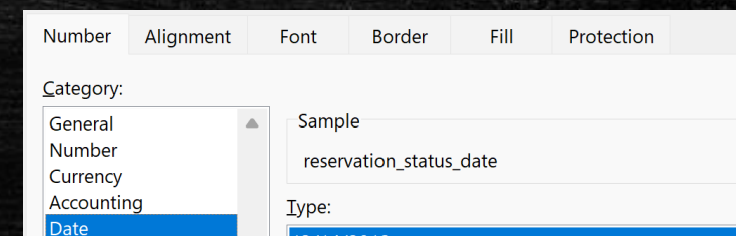
Task 1

Phase 1: Data Cleaning & Preparation

- Start with data Importing and Data Exploration and understanding each column and its values mean
- Remove duplicates (if any) In Primary Key(Booking_ID)
- Handle missing values
- Standardize text fields
- Identify Arrival Date Issue



Arrival_Date



```
=DATE(@arrival_date_year,MONTH(DATEVALUE(1&@arrival_date_month)),@arrival_date_day_of_month)
```


Task 2

Phase 2: Creating useful (Calculated Columns)

- Total Stay Nights
- Guest Type ("Single", "Couple", "Family")
- Arrival date (Month/ Day/ Year)
- Payment Risk("High Risk", "Low Risk") Based on Deposit Status
- Total Guests

Total Stay Nights ▼

=@stays_in_weekend_nights+@stays_in_week_nights

Guest Type ▼

=IF(@children+@babies>0,"Family",IF(@adults=1,"Single","Couple"))

Arrival_Date ▼

=DATE(@arrival_date_year,MONTH(DATEVALUE(1&@arrival_date_month)),@arrival_date_day_of_month)

Payment Risk ▼

=IF(@deposit_type="No Deposit","High Risk","Low Risk")

Total Guests ▼

=@adults+@children+@babies

Task 2

Phase 2: Creating useful (Calculated Columns)

- Revenue per Booking
- Booking Status (Cancelled/ Not Cancelled)
- Lead Time Category (Long: Higher than 90 days , Medium, Short: less than 30)
- Feel Free to create more calculated columns to enhance your Analysis

Season

=SWITCH(@arrival_date_month,
"December","Winter",
"January","Winter",
"February","Winter",
"March","Spring",
"April","Spring",
"May","Spring",
"June","Summer",
"July","Summer",
"August","Summer",
"September","Autumn",
"October","Autumn",
"November","Autumn")

Revenue per Booking

=@Total_Stay_Nights*@Average_Daily_Rate

Booking Status

=IF(@is_canceled=1,"Cancelled","Not Cancelled")

Lead Time Category

=IF(@lead_time>90,"Long",IF(@lead_time<30,"Short","Medium"))

Revenue Category

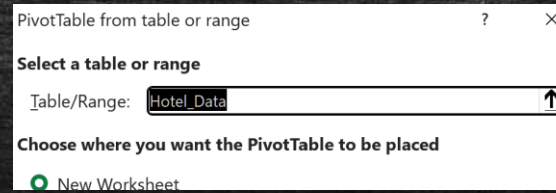
=IF(@Revenue_per_Booking>1500,"High",IF(@Revenue_per_Booking<500,"Low","Medium"))

Stay Type

=IF(@Total_Stay_Nights>5,"Long Stay",IF(@Total_Stay_Nights<3,"Short Stay","Medium Stay"))

Task 3

Phase 3: PivotTables & Insights



- Ask Strong business Questions and answer it by Pivot Tables

Question 1					Question 2						Question 3				
Which hotel type generates the highest revenue?					Which month has the highest number of bookings?					Which customer type has the highest cancellation rate?					
City Hotel					August					Transient					

Question 4						Question 5						Question 6						
Which country generates the highest number of bookings?						Which market segment generates the highest revenue?						Does lead time affect cancellations?						
PRT						Online TA						Yes. As lead time increases, the likelihood of cancellation also increases.						

Question 7				Question 8				Question 9			
Which guest type books the most?				Which season has the highest revenue?				What is the hotel booking cancellation rate?			
Couple				Summer				37.04%			

Task 4

Phase 4: Data Visualization

- Create charts to Visualize each
- Dashboard Creation: Build an interactive dashboard including:

KPIs:

1. Total Bookings
2. Total Revenue
3. Cancellation Rate

Total Revenue	Total Booking	Cancellation Rate
\$42,723,498	119,390	37.04%

Task 4

Phase 4: Data Visualization

- Create charts to Visualize each
- Dashboard Creation: Build an interactive dashboard including:

Filters (Slicers):

1. Hotel Type
2. Year
3. Customer Type

- Charts & PivotTables

arrival_date_year	≡	✕
2015		
2016		
2017		

hotel	≡	✕
City Hotel		
Resort Hotel		

customer_type	≡	✕
Contract		
Group		
Transient		
Transient-Party		

Dashboard

Hotel Performance
Analysis



Analysis



Data



Filter Panel

arrival_date_year



2015
2016
2017

hotel



City Hotel
Resort Hotel

customer_type



Contract
Group
Transient
Transient-Party

Total Revenue

\$42,723,498

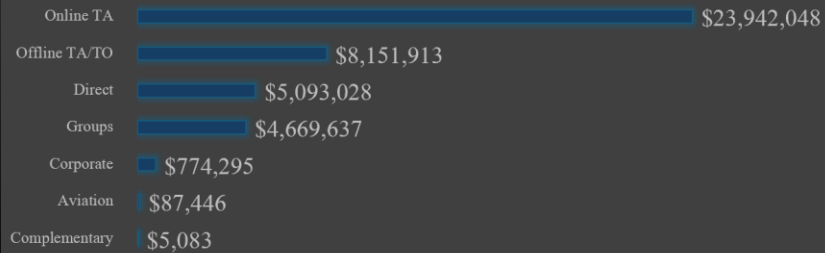
Total Booking

119,390

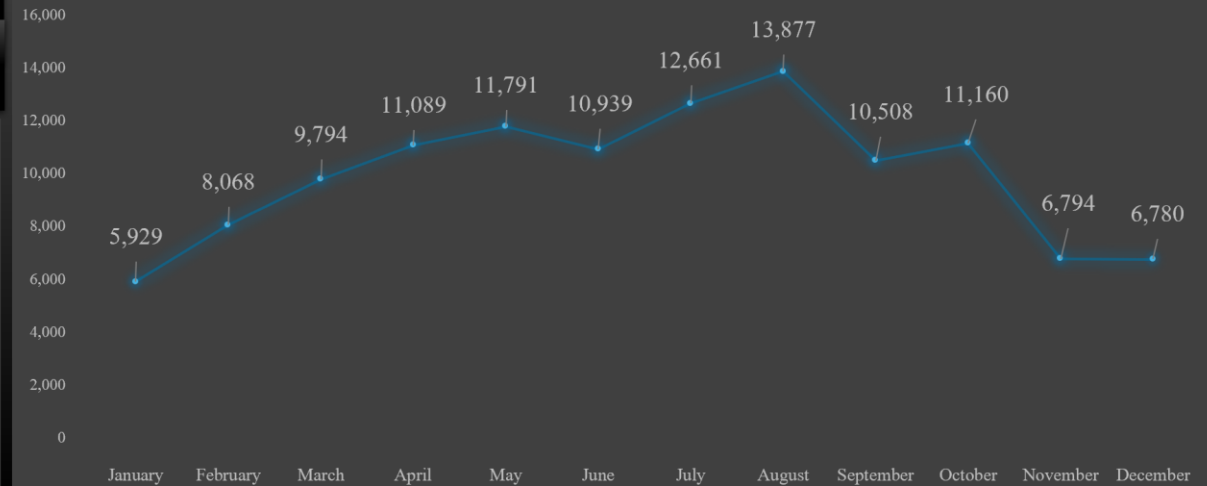
Cancellation Rate

37.04%

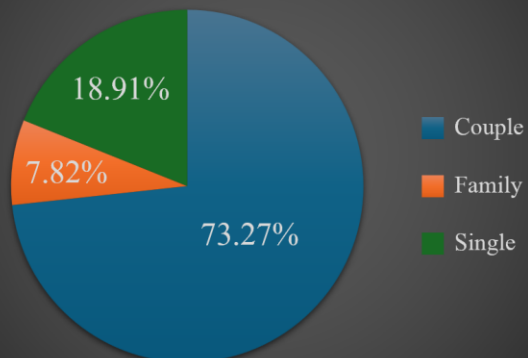
Revenue by Market Segment



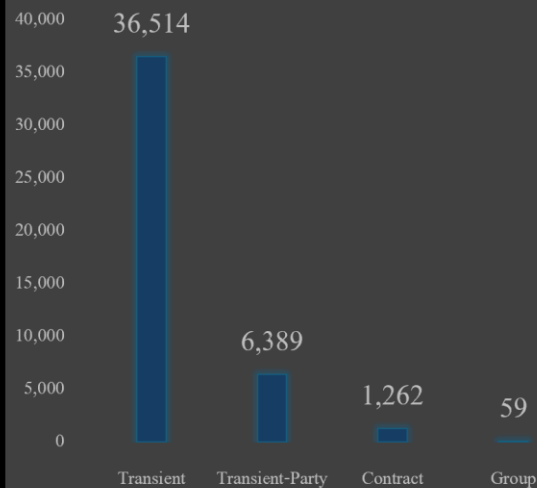
Bookings by Month



Booking by Guest Type



Cancellation by Customer Type



Monthly Bookings & Revenue

